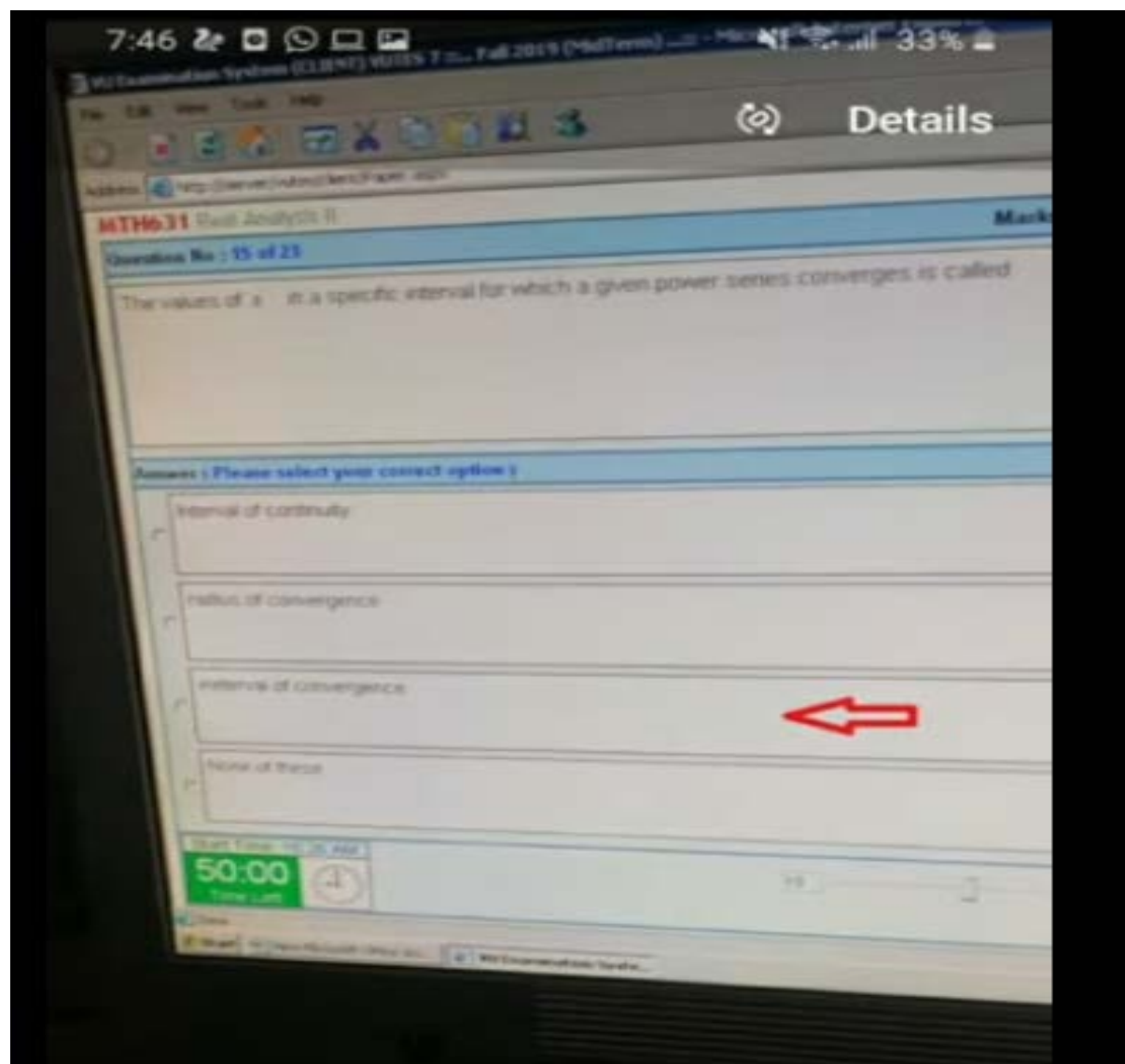
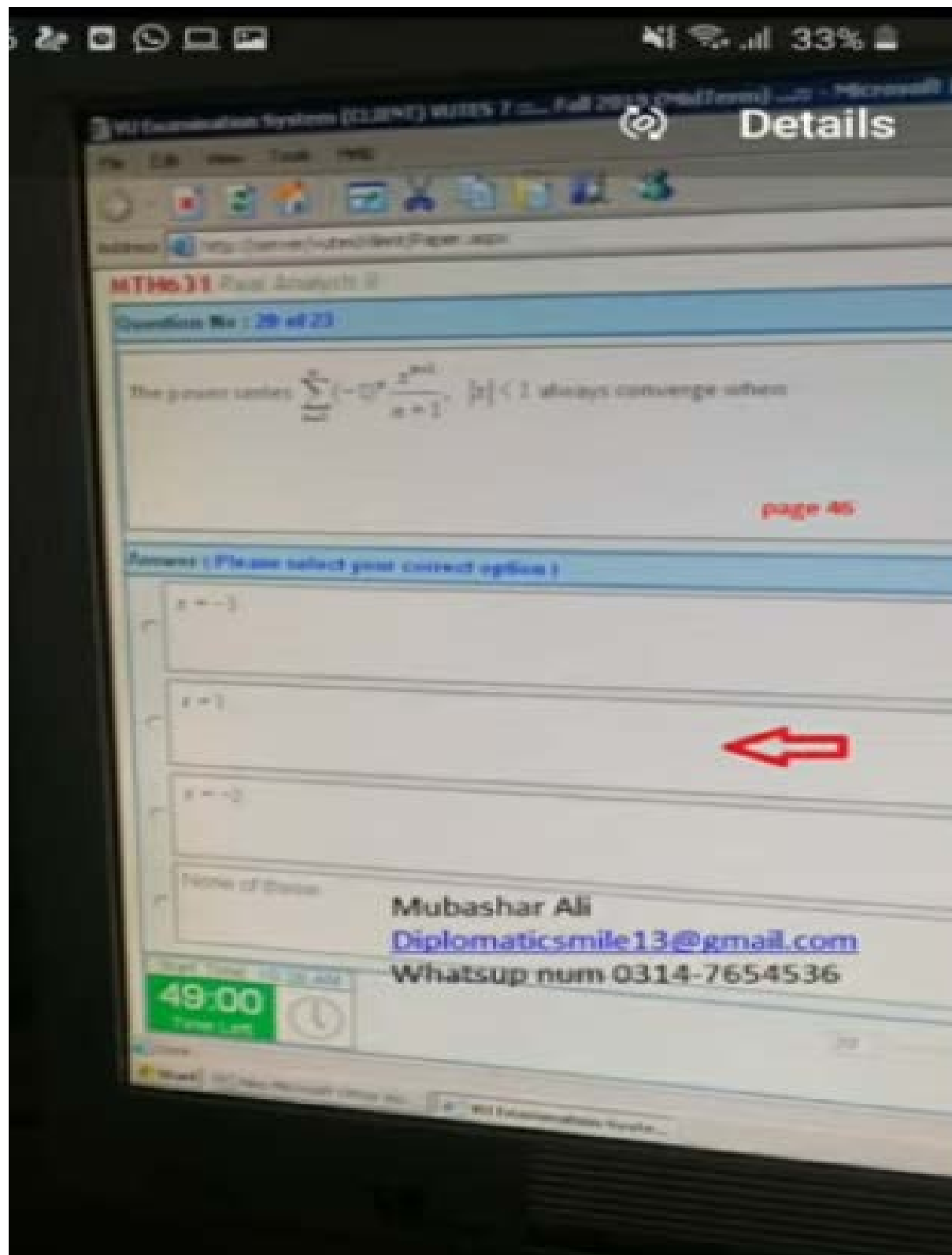


MTH631 FINAL PAPER





MTH631 Real Analysis II

Questions No : 18 of 23

A set of points of the form $Z = Z_0 + iU$, $t_1 \leq t \leq t_2$ is called _____

page 75

Answers (Please select your correct option)

☐ $N_r(Z_0) = \{Z \mid \|Z - Z_0\| = r\}$

☐ $N_r(Z_0) = \{Z \mid \|Z - Z_0\| > r\}$

☐ $N_r(Z_0) = \{Z \mid \|Z - Z_0\| < r\}$

☐ None of these

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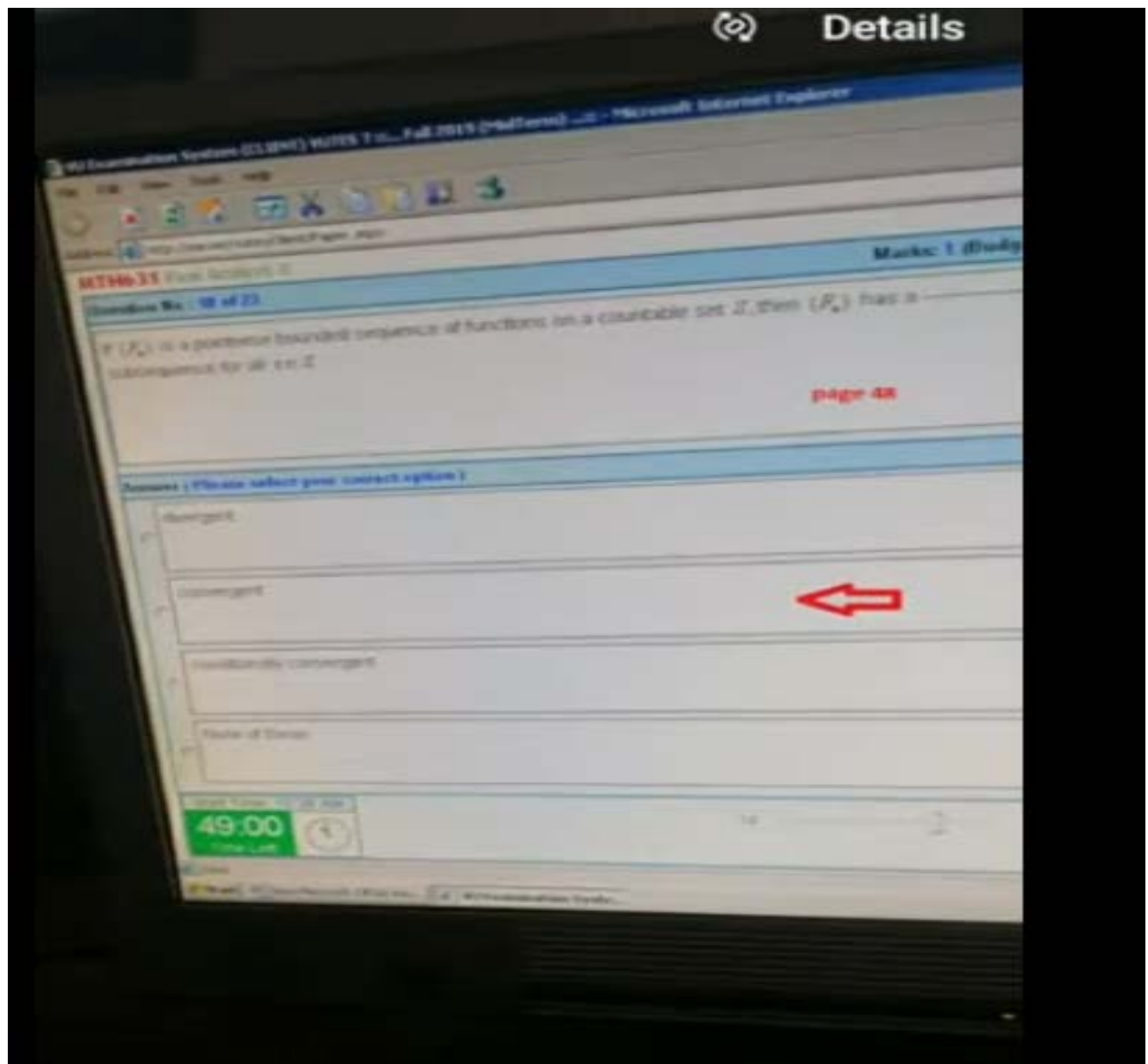
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MTH631 Real Analysis II

Question No: 17 of 23

For the series $f(x) = \sum_{n=0}^{\infty} (-1)^n x^n = \frac{1}{1+x}$, the $\lim_{x \rightarrow 0} f(x) =$

page 46

Answer: (Please select your correct option)

- ☐ $\frac{1}{2}$
- ☐ $\frac{1}{3}$
- ☐ $\frac{1}{4}$
- ☐ None of these



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RTM6.31

Question No. 16.4.22

Marked 1 (Budgeted Time 1.1)

If $\{f_n\}$ is a sequence of continuous functions defined on a compact set X and f is a continuous bounded and continuous on X , then:

page 50

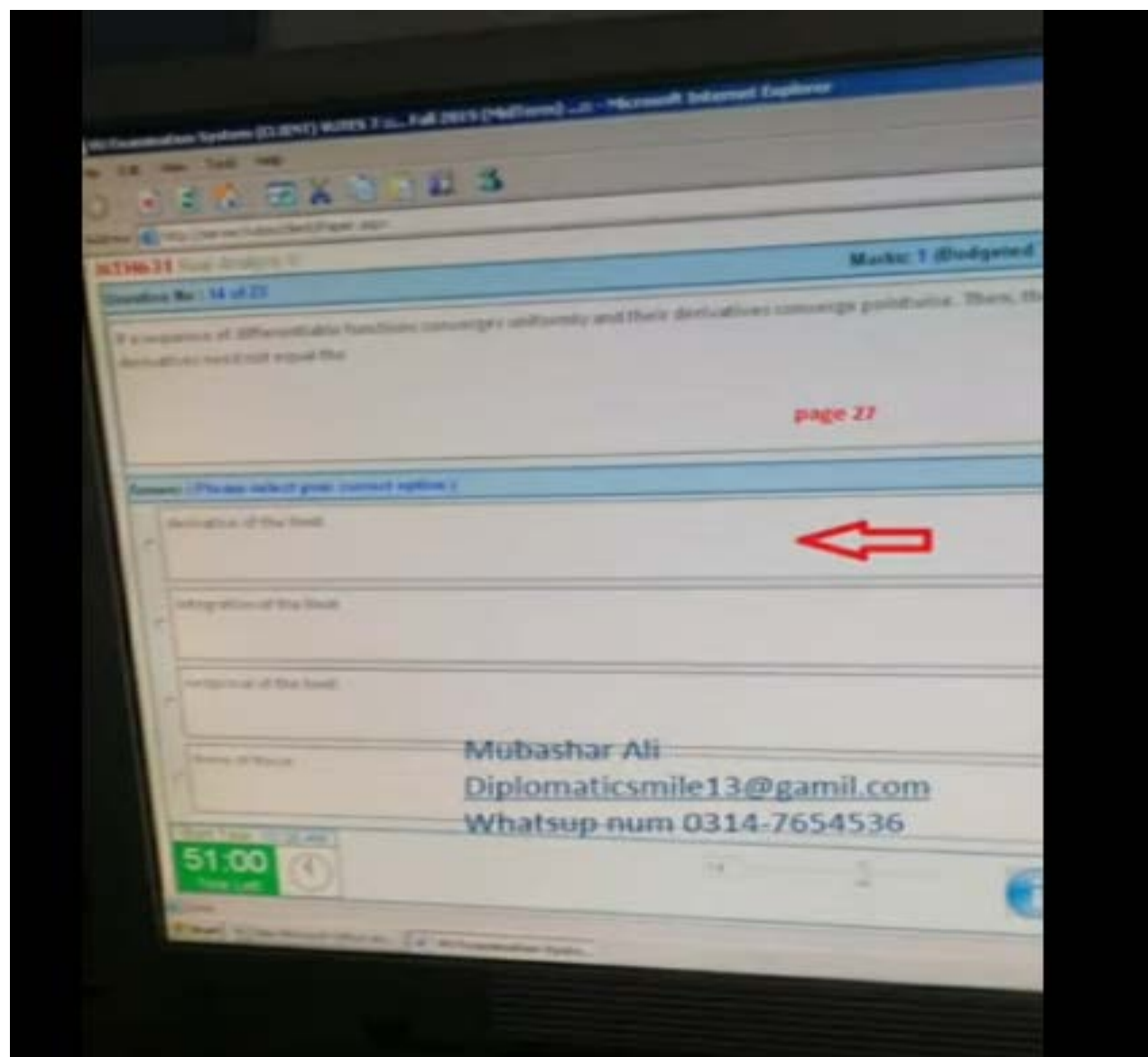
Answer: Please select your correct option(s)

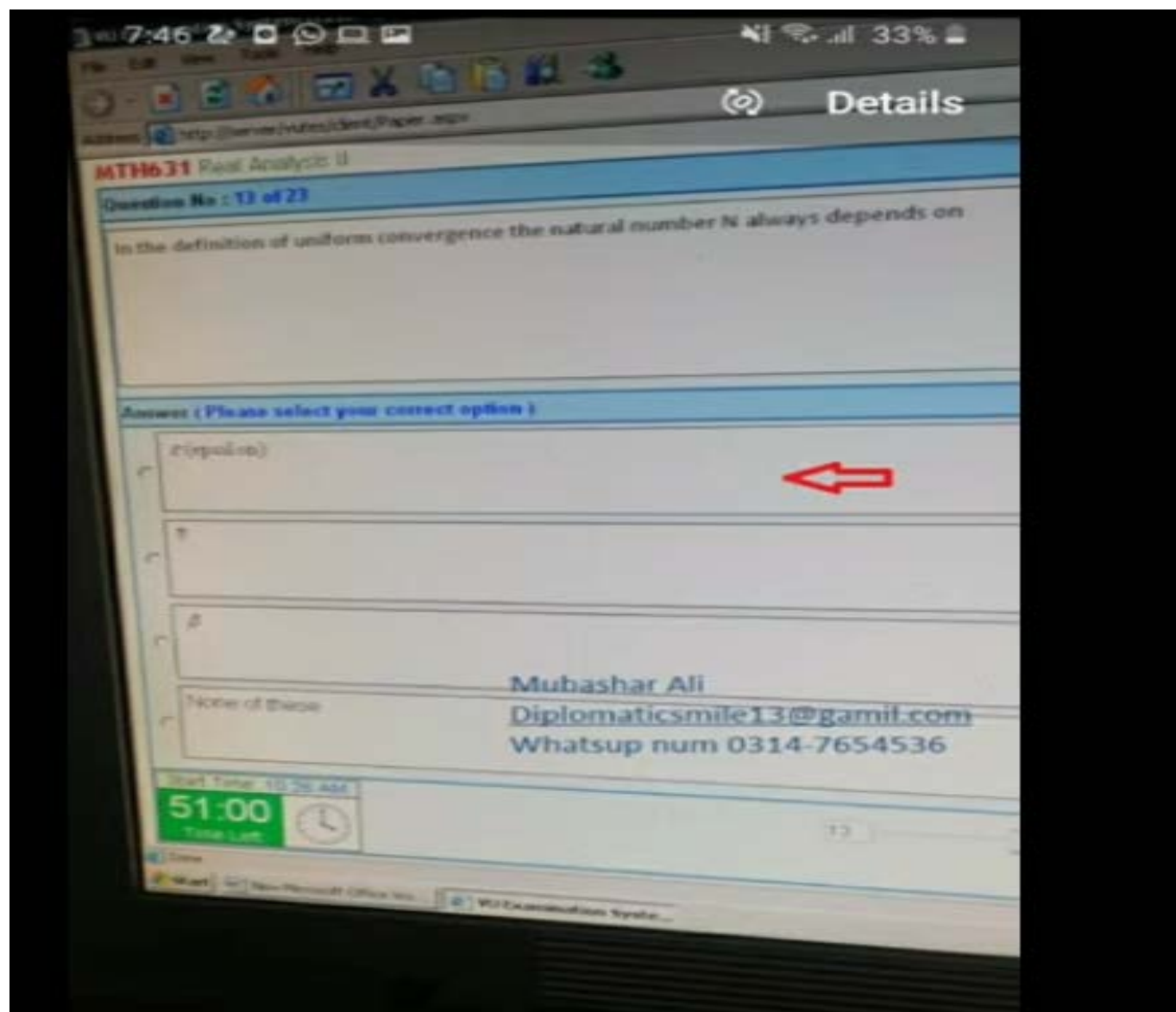
- ☐ $\{f_n\}$ contains a uniformly convergent subsequence.
- ☐ $\{f_n\}$ does not contain a uniformly convergent subsequence.
- ☐ $\{f_n\}$ contains a pointwise convergent subsequence.

Marked 1 (Budgeted Time 1.1)

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MT99631 Real Analysis II

Question No: 52 of 23

If we differentiate the series $\sum_{n=1}^{\infty} c_n x^n$ term by term, we get the series of the form

page 30

Answer (Please select your correct option)

☐ $-\sum_{n=1}^{\infty} \frac{1}{n} c_n x^{n-1}$

☐ $-\sum_{n=1}^{\infty} \frac{1}{n} c_n x^n$

☐ $-\sum_{n=1}^{\infty} \frac{1}{n} c_n x^n$

☐ None of these

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MTH631 Real Analysis II

Question No: 11 of 20

The limit of a uniformly convergent series of integrable functions is always

page 25

Answer (Please select your correct option)

- ☐ integrable
- ☐ not integrable
- ☐ differentiable
- ☐ None of these



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MTH631 Real Analysis II

Question No : 10 of 23

If a series converges absolutely uniformly on S , then

page 20

Answer (Please select your correct option)

☐ it converges uniformly on S



☐ it diverges

☐ it is differentiable on S

☐ None of these

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MTH631 Real Analysis II



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Question No : 9 of 23

Which of the following is true about the expression $\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x)dx$?

page 11

Answer (Please select your correct option)

☐ It is always true☒ It is not true generally☐ It is always false☐ None of these

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MTH631 Real Analysis I

Question No: 8 of 23

Suppose that a sequence of function (f_n) is pointwise convergent on the set S . Then

Answer: (Please select your correct option.)

☐ (f_n) always converges uniformly



☐ (f_n) may converge uniformly

☐ (f_n) always diverges

☐ None of these

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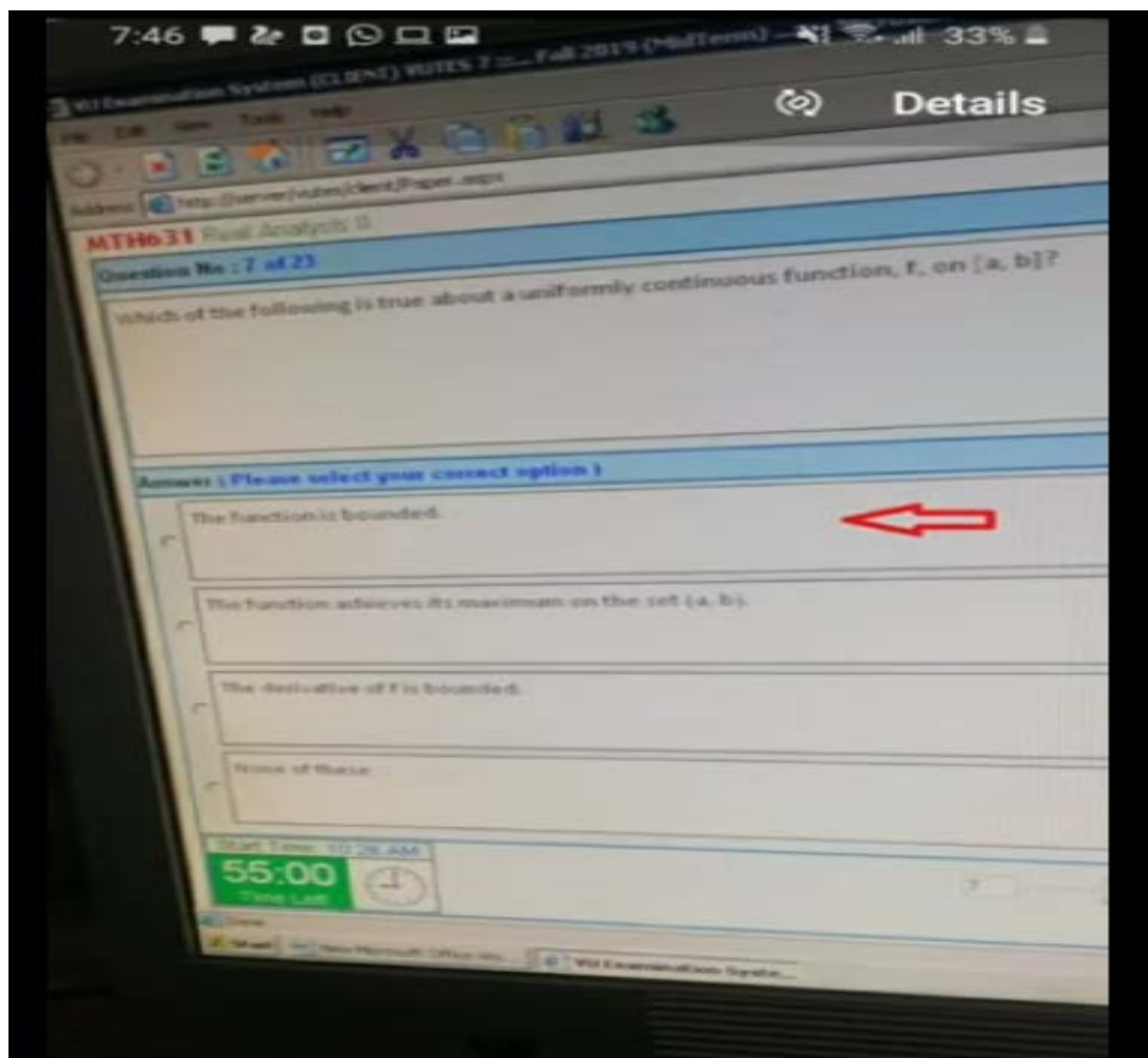
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MTH631 Real Analysis II



Details

Question No : 6 of 21

If Dirichlet conditions are satisfied, then the Fourier series always

Answer (Please select your correct option)

☐ has no conclusion

☐ None of these

☐ diverges

☐ converges

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MTH631 Real Analysis II



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Question No : 5 of 23

which property is not satisfied by norm defined on a set S .

page 5

Answer (Please select your correct option)

☐ $\{x + A_S\} \subseteq \{y + A_S\}$

☐ $\{y + A_S\} \subseteq \{x + A_S\}$

☐ $\{x - A_S\} \subseteq \{y - A_S\}$

☐ None of these

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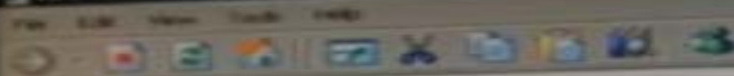
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MTH631 Real Analysis II

Question No : 4 of 23

Which statement(s) is (are) true about the sequence of functions defined by

$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} \infty, & x < 0, \\ 1, & x = 0, \\ 0, & 0 < x \leq 1 \end{cases}$$

page 3

Answer (Please select your correct option)

- ☐ The set on which sequence of functions is defined is $D = [-5, 5]$
- ☐ The sequence of functions is pointwise convergent to $f(x) = 0$
- ☐ The sequence converges pointwise on $D = [0, 1]$ to the limit function f defined by $f(x)$
- ☐ None of these

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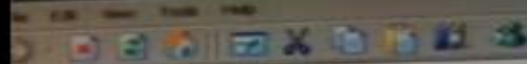


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MT19631 Real Analysis II

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Question No : 3 of 23

By Stone and Weierstrass theorem, the trigonometric polynomials are dense in $C[a,b]$ provide

page 56 handout

Answer (Please select your correct option)

☐ $b-a < 2\pi$



☐ $b+a < 2\pi$

☐ $b-a > 2\pi$

☐ None of these

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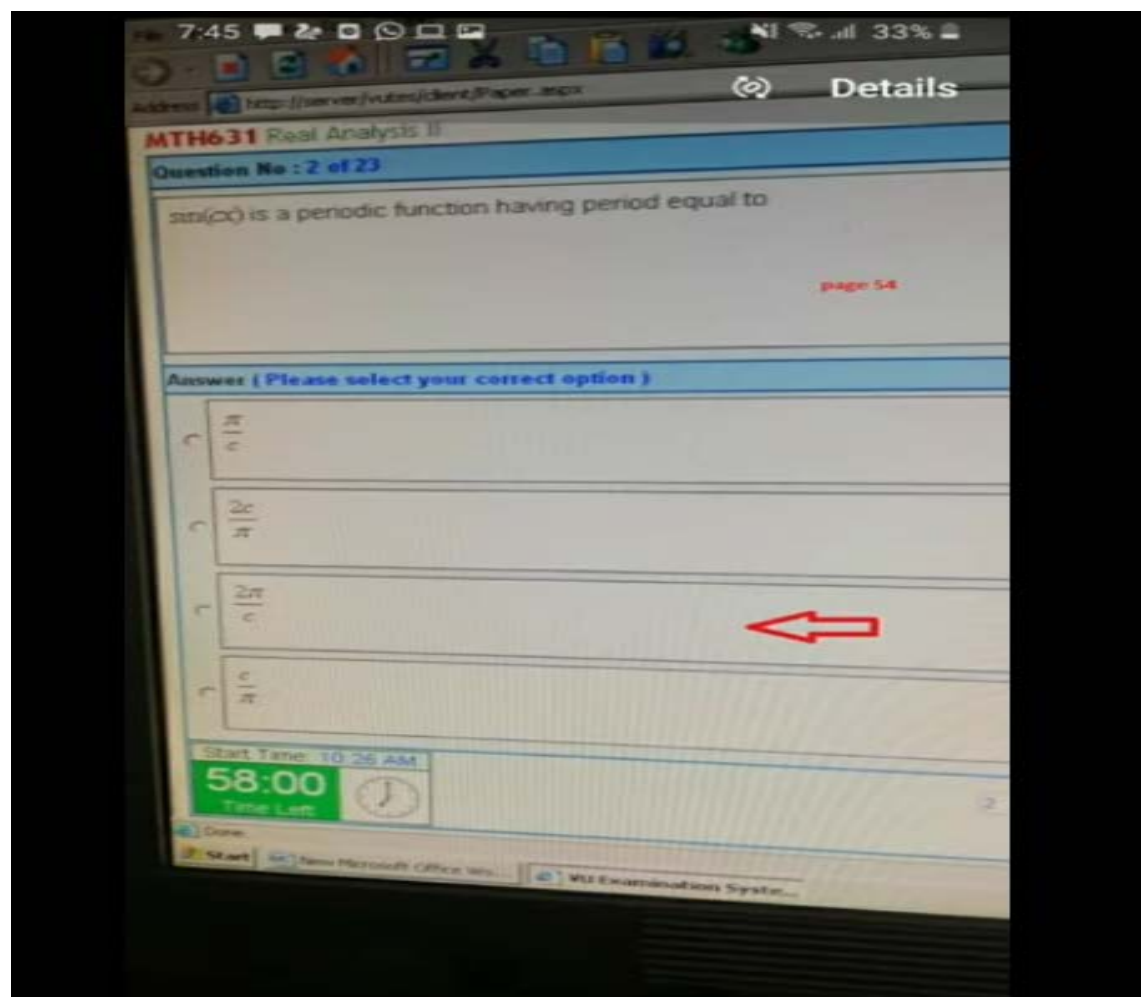
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MTH631

Question No : 1 of 23

An infinite series of the form $\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$ is called of f about x_0 .

page 16

Answer (Please select your correct option)

☐ Power series

☐ Maclaurin series

☐ Taylor series

☐ None of these

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MTH631 Real Analysis II

Question No : 20 of 25

If f and g are integrable on S and $f(X) \leq g(X)$ for X in S , then ———

[page 163](#)

Answer (Please select your correct option)

☐ $\int_S f(X) dX \geq \int_S g(X) dX$

☒ $\int_S f(X) dX \leq \int_S g(X) dX$

☐ $\int_S f(X) dX > \int_S g(X) dX$

☐ None of these

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The lower integral $\int_a^b f(x)dx$ of f over R is the supremum of all the _____.

page 155

Answer (Please select your correct option)

☐ lower sums



☐ upper sums

☐ average sums

☐ None of these

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The upper integral $\overline{\int}_R f(X)d(X)$ of f over R is the infimum of all the _____

[page 135](#)

Answer (Please select your correct option)

☐ lower sums

☐ upper sums

☐ average sums

☐ None of these

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A function f is said to be absolutely integrable on $[a, b)$ if f is locally integrable on $[a, b)$ and ———

page 145

Answer (Please select your correct option)

☐ $\int_a^b |f(x)| dx < \infty$



☐ $\int_a^b |f(x)| dx = \infty$

☐ $\int_a^b |f(x)| dx \geq \infty$

☐ None of these

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If f and g are locally integrable on (a, b) , $g(x) > 0$ and $f(x) \geq 0$ on some subintervals (a, b) of $[a, b]$, and $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = M$ if $M = 0$ and $\int_a^b g(x) dx < \infty$, then

page 144

Answer (Please select your correct option)

☐ $\int_a^b f(x) dx$ and $\int_a^b g(x) dx$ converges together

☐ $\int_a^b f(x) dx = \infty$

☒ $\int_a^b f(x) dx < \infty$

☐ $\int_a^b f(x) dx \geq \infty$

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Question Summary: 1 Attempted Question

The function $f(x) = (1-x)^{-p}$ is locally integrable on _____

page 139

Answer (Please select your correct option)

☐ [0,1]

☐ [-1,1]

☐ (0,1)

☒ [0,1]

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If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then p is called ———

[page 134](#)

Answer (Please select your correct option)

☐ positive semidefinite

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☐ positive definite

☒ negative semidefinite



☐ negative definite

Question No : 13 of 26

If value of a becomes zero in the Taylor series $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$, then the series is called ———

[page 44](#)

Answer (Please select your correct option)

☐ Maclaurin series



☐ Fourier series

☐ Power series

☐ None of these

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MTH631 Real Analysis II

Question No : 12 of 26

If $h = f(r, \theta)$, then using the chain rule, $\frac{\partial h}{\partial r} =$ _____

ans.127

Answer (Please select your correct option)

☐ $\frac{\partial f}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial x}$

☐ $\frac{\partial f}{\partial x} + \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial r}$

☐ $\frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r}$



☐ None of these

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8

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If $f(x) = 3x + 5x^3$, then at $x = -1$, $df =$ _____

Answer (Please select your correct option)

☐ $18dy$

☐ $18dx$

☐ 18



MTH631 Real Analysis II

Question No : 19 of 26

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \dots + m_n x_n$ for some constants m_i 's is called ———

[page 112](#)

Answer (Please select your correct option)

☐ linear function



☐ quadratic function

☐ constant function

☐ None of these

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MTH631 Real Analysis II

Question No : 9 of 26

For the function $f(x,y) = 3x^2y^3 + xy$ in two variables, $f_{xy}(0,0) = \dots\dots\dots$

Answer (Please select your correct option)

☐

1

☐

-1

☐

0

☐

None of these



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MTH631 Real Analysis II

Question No : 8 of 26

A critical point at which a function attains its minimum value among all points where it is defined is called

Answer (Please select your correct option)

☐ global maximum

☒ global minimum

☐ supremum

☐ None of these

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A $f_1, f_2, \dots, f_n$ exist on a neighborhood of X_0 and are continuous at X_0 , then f is ——— at X_0 .

[page 115](#)

Answer (Please select your correct option)

☐ uniformly converget

☐ piecewise convergent

☐ differentiable

☐ None of these

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Question No : 6 of 26

A set $A \subset \mathbb{R}$ of real numbers is _____ if there exists a real number $M \in \mathbb{R}$, such that $x \leq M$ for every $x \in A$.

Answer (Please select your correct option)

☐ bounded below



☐ uniformly continuous

☐ bounded above

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☐ None of these

In a vector valued function $G = (g_1, g_2, \dots, g_n)$. Then $g_1, g_2, \dots, g_n$ are called the _____ functions of G .

page 114

Answer (Please select your correct option)

☐ bijective

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☐ surjective

☐ component



☐ None of these

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MTH631 Real Analysis II

Question No : 4 of 26

The quotient of two continuous functions is a _____ function wherever the denominator is non-zero.

Answer (Please select your correct option)

☐ differentiable

☒ continuous

☐ composite

☐ None of these



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Question No : 2 of 26

Any polygonally connected set either it is open or not, is always _____

page 82

Answer (Please select your correct option)

☐ connected

☐ disconnected

☐ a region



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The set $S = \{(x, y), x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 \geq 4\}$ is not a region in $\mathbb{R}^2$, since it is _____

page 90

Answer (Please select your correct option)

☐ not connected.



☐ connected.

☐ polygonally connected.

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☐ None of these

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Question No : 2 of 26

A set S is polygonally connected if every pair of points in S can be connected by a polygonal path lying _____

[page 89](#)

Answer (Please select your correct option)

☐ outside the set S

☐ partially in S

☒ entirely in S

☐ None of these

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A set is closed if and only if it always contains all of its _____

page 86

Answer (Please select your correct option)

☐ limit points



☐ neighbourhood points

☐ interior points

☐ None of these

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MTH631 Real Analysis II

Question No : 26 of 26

Suppose that $f(x, y) = \begin{cases} 0 & \text{if } x \text{ and } y \text{ are rational,} \\ 1 & \text{if } x \text{ is rational and } y \text{ is irrational,} \\ 2 & \text{if } x \text{ is irrational and } y \text{ is rational,} \\ 3 & \text{if } x \text{ and } y \text{ are irrational.} \end{cases}$

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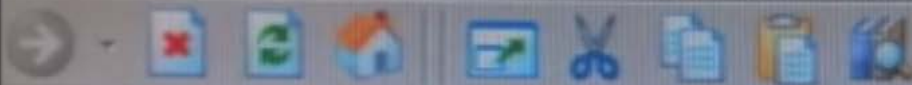
$$\overline{\int}_R f(x, y) d(x, y) \quad \text{and} \quad \underline{\int}_R f(x, y) d(x, y) \quad \text{if} \quad R = [a, b] \times [c, d].$$

Let $P = \{R_1, R_2, \dots, R_k\}$. Let

$$M_j = \sup \{f(x, y) \mid X \in R_j\} \quad \text{and} \quad m_j = \inf \{f(x, y) \mid X \in R_j\}.$$

Let j be arbitrary in $\{1, 2, \dots, k\}$. Since R_j contains a point $(\hat{x}_j, \hat{y}_j)$ with $\hat{x}_j$ and $\hat{y}_j$ irrational, $M_j = 3$. Hence, $\overline{\int}_R f(x, y) d(x, y) = 3(b-a)(d-c)$. Since R_j contains a point $(\tilde{x}_j, \tilde{y}_j)$ with $\tilde{x}_j$ and $\tilde{y}_j$ rational, $m_j = 0$. Hence, $\underline{\int}_R f(x, y) d(x, y) = 0$.

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MTH631 Real Analysis II

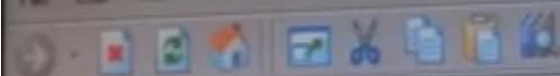
Question No : 25 of 26

Prove: If a nonempty subset of $\mathbb{R}^n$ is both open and closed, then $S = \mathbb{R}^n$.

Suppose that X_0 is in S . If $S \neq \mathbb{R}^n$, there is an $X_1 \notin S$. Let

$$H = \{ \tau \mid (1 - \tau)X_0 + \tau X_1 \in S \text{ for } 0 \leq \tau < 1 \}.$$

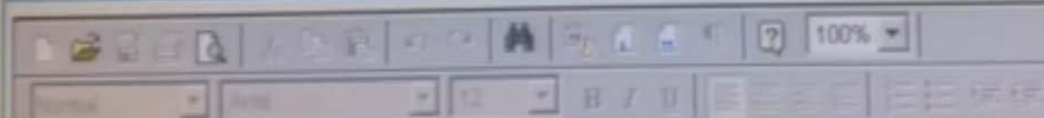
Since S is open it contains a neighborhood of X_0 , so $H \neq \emptyset$. Since X_1 is not in S , $\tau \leq 1$ for all τ in H . Let $\rho = \sup H$ and $\bar{X} = (1 - \rho)X_0 + \rho X_1$; then $\bar{X}$ is a limit point of S and so in S , because S is closed. Since S is open, it contains some ϵ -neighborhood of $\bar{X}$, so $\rho + \epsilon/2$ is in H . This contradicts the definition of ρ . Hence, $S = \mathbb{R}^n$.

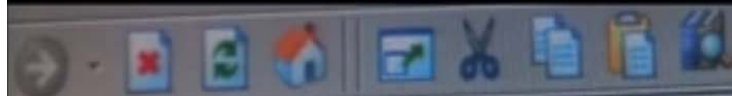
**MTH631** Real Analysis II

Question No : 24 of 26

If R is degenerate, then prove by using the definition of Riemann integral that $\int_R f(\mathbf{X}) d\mathbf{A} = 0$ if f is bounded on R .

Suppose, for example, that $R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$ with $a_1 = b_1$. Then every partition $\mathbf{P}$ of R is of the form $\mathbf{P} = \{R_1, R_2, \dots, R_k\}$, where $v(R_j) = 0$, $1 \leq j \leq k$. Therefore, every Riemann sum of f over R equals zero, so $\int_R f(\mathbf{X}) d\mathbf{X} = 0$.

Answer ([Please click here to Add Answer](#))



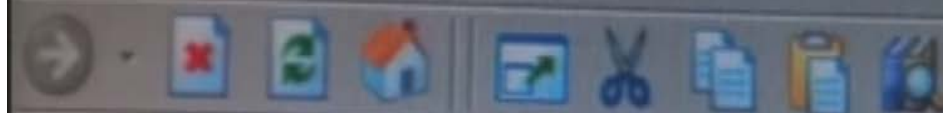
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MTH631 Real Analysis II

Question No : 23 of 26

Determine whether the integral $\int_0^{\infty} \frac{1 + \cos^2 x}{\sqrt{1 + x^2}} dx$ converges or diverges.

$$\frac{1 + \cos^2 x}{\sqrt{1 + x^2}} > \frac{1}{\sqrt{1 + x^2}} = \frac{1}{x\sqrt{1 + 1/x^2}} > \frac{1}{x\sqrt{2}} \text{ if } x > 1. \text{ Since } \int_1^{\infty} \frac{dx}{x} = \infty,$$
$$\int_0^{\infty} \frac{1 + \cos^2 x}{\sqrt{1 + x^2}} dx = \infty$$



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MTH631 Real Analysis II

Question No : 22 of 26

Calculate $\frac{\partial f(X)}{\partial \Phi}$ where $f(x, y, z) = \log(1 + x + y + z)$, $\Phi = (0, 1, 0)$.

$$\begin{aligned} h(t) &= f(x + \phi_1 t, y + \phi_2 t, z + \phi_3 t) = \log(1 + x + y + z + (\phi_1 + \phi_2 + \phi_3)t); h'(t) = \\ &\frac{\phi_1 + \phi_2 + \phi_3}{1 + x + y + z + (\phi_1 + \phi_2 + \phi_3)t}; \frac{\partial f(X)}{\partial \dot{L}} = h'(0) = \frac{\phi_1 + \phi_2 + \phi_3}{1 + x + y + z}; \text{ if } \dot{L} = (0, 1, 0) \\ \text{then } \frac{\partial f(X)}{\partial \dot{L}} &= \frac{1}{1 + x + y + z}. \end{aligned}$$

Question No : 20 of 26

A _____ surface in $\mathbb{R}^n$ has zero content.

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Answer (Please select your correct option)

☐ parabolic

☐ elliptical

☐ differentiable

☐ None of these

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Question No : 13 of 26

If f is integrable on a rectangle R , then ———[page 158](#)

Answer (Please select your correct option)

☐ $\int_a^b f(x) dx > \int_a^b f(x) dx = \int_a^b f(x) dx$

☒ $\int_a^b f(x) dx = \int_a^b f(x) dx = \int_a^b f(x) dx$

☐ $\int_a^b f(x) dx > \int_a^b f(x) dx > \int_a^b f(x) dx$

☐ None of these

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The upper integral $\int_a^b f(x) dx$ of f over R is the infimum of all the _____

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Answer (Please select your correct option)

☐ lower sums☐ upper sums☐ average sums☐ None of these

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f and g are locally integrable on (a, b) , $g(x) > 0$ and $f(x) \geq 0$ on some subintervals $[a_1, b_1]$ of (a, b) , and $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = M$, if $M = \infty$ and $\int_{a_1}^{b_1} g(x) dx = \infty$,

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Answer (Please select your correct option)

☐ $\int_a^b f(x) dx$ and $\int_a^b g(x) dx$ converges together

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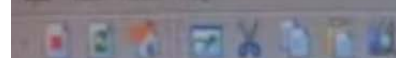
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☐ $\int_a^b f(x) dx \geq \infty$

☒ $\int_a^b f(x) dx = \infty$



☐ $\int_a^b f(x) dx < \infty$



If f and g are locally integrable on $[a, b]$, $g(x) > 0$ and $f(x) \geq 0$ on some subintervals $[a_1, b]$ of $[a, b]$, and $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = M$. If $M = 0$ and $\int_a^b g(x) dx < \infty$, then

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Answer (Please select your correct option)

☐ $\int_a^b f(x) dx$ and $\int_a^b g(x) dx$ converges together

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☐ $\int_a^b f(x) dx = \infty$

☐ $\int_a^b f(x) dx < \infty$



☐ $\int_a^b f(x) dx \geq \infty$

If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$ & $p(X) < 0$ for all $X \neq X_0$. Then p is called ———

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Answer (Please select your correct option)

☐ positive semidefinite

☐ positive definite

☐ negative semidefinite

☐ negative definite

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Question No : 14 of 26

If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \geq 0$ for all X , then p is called _____

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Answer (Please select your correct option)

☐ positive semidefinite

☐ positive definite

☐ negative semidefinite

☐ negative definite

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Question No : 13 of 26

If value of a becomes zero in the Taylor series $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$, then the series is called ———

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Answer (Please select your correct option)

☐ Maclaurin series



☐ Fourier series

☐ Power series

☐ None of these

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If $h = f(x, y)$, $x = f(r, \theta)$ and $y = f(r, \theta)$, then using the chain rule, $\frac{\partial h}{\partial \theta} =$ _____

page 127

Answer (Please select your correct option)

☐ $\frac{\partial f}{\partial r} \frac{\partial r}{\partial \theta} + \frac{\partial f}{\partial r} \frac{\partial r}{\partial \theta}$

☐ $\frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta}$

☐ $\frac{\partial f}{\partial x} + \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial \theta}$



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Question No : 11 of 26

If $f(x) = 3x + 5x^3$, then at $x = -1$, $df =$ _____

Answer (Please select your correct option)

☐ $18dy$

☐ $18dx$

☐ 18

☐ None of these

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A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \dots + m_n x_n$ for some constants m_i 's is called ———

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Answer (Please select your correct option)

☐ linear function

☐ quadratic function

☐ constant function

☐ None of these



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MTH631 Real Analysis II

Question No : 9 of 26

The rate of change of $f(x, y)$ in the direction of a unit vector is called the _____ of $f(x, y)$

Answer (Please select your correct option)

☐ differential

☒ directional derivative

☐ mixed derivative

☐ None of these



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MTH631 Real Analysis II

Question No : 8 of 26

A critical point at which a function attains its minimum value among all points where it is defined is called

Answer (Please select your correct option)

☐ global maximum

☒ global minimum

☐ supremum

☐ None of these



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MTH631 Real Analysis II

Question No : 7 of 26

Marks: 1 (B)

A function f is _____ on a subset S of $\mathbb{R}^n$ if S is contained in an open set on which $f, f_x, f_{x_2}, \dots, f_{x_n}$ are continuous.

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Answer (Please select your correct option)

☐ continuously differentiable



☐ uniformly convergent

☐ piecewise convergent

☐ None of these

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Question No : 6 of 26

A set $A \subset \mathbb{R}$ of real numbers is _____ if there exists a real number $M \in \mathbb{R}$, such that $x \leq M$ for every $x \in A$.

Answer (Please select your correct option)

☐ bounded below

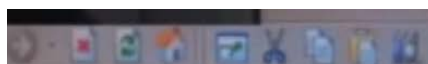
☐ uniformly continuous

☐ bounded above

☐ None of these



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MTH631 Real Analysis II

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Question No: 5 of 26

Marks: 1 (Budgeted Time: 10)

Suppose $U_0 \in T$ is a limit point of T . g is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then _____ is continuous at U_0 .

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Answer (Please select your correct option)

☐ $f \circ g$



☐ $\frac{f}{g}$

☐ $\frac{g}{f}$

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☐ None of these



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MTH631 Real Analysis II

Question No : 4 of 25

Marks: 1

If f and g are continuous functions at the point X_0 in $\mathbb{R}^n$, then $\frac{f}{g}$ is continuous at X_0 provided that _____

page 100

Answer (Please select your correct option)

☐ $g(X_0) = 0$

☒ $g(X_0) \neq 0$

☐ $f(X_0) = g(X_0) = 0$

☐ None of these

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Question No : 3 of 26

For the function $f(x,y) = \frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y) \rightarrow (0,0)$ along the line $y=x$ is —

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Answer (Please select your correct option)

☐ $\frac{1}{2}$ ☐ $\frac{1}{\sqrt{2}}$ ☐ 0☐ 1

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